

COPD Exacerbation — Inpatient Management

Bronchodilators + steroids + targeted antibiotics + the right oxygen device · NIV is the single biggest mortality lever · treat the cause, set up the discharge to prevent the next one

1 · Define & triage

An exacerbation is an acute worsening of dyspnea, cough, and/or sputum (volume or purulence) beyond day-to-day variation. **GOLD 2026** grades severity by measured physiology, not gestalt.

Severity	Rough markers
Mild	Short-acting bronchodilators only
Moderate	SABA + steroids ± antibiotics
Severe	ED/hospital; possible respiratory failure

Admit for: marked symptom increase (rest dyspnea), severe underlying COPD, failure of outpatient therapy, new hypoxemia/hypercapnia, altered mentation, or inadequate home support.

2 · The core bundle

- **Bronchodilators:** short-acting beta-agonist ± short-acting antimuscarinic, scheduled + PRN (nebs or MDI-spacer equivalent).
- **Systemic steroids:** prednisone **40 mg PO × 5 days** (REDUCE) — oral equals IV; no taper needed for a 5-day course.
- **Antibiotics** if ≥ 2 of: ↑dyspnea, ↑sputum volume, ↑sputum purulence (purulence is the key one) — or if mechanically ventilated. Usual 5 days; cover H. flu/pneumococcus/Moraxella (e.g., macrolide, doxycycline, or amox-clav); add Pseudomonas coverage if risk factors.
- **Oxygen:** titrate to SpO₂ **88–92%** — avoid hyperoxia, which worsens hypercapnia.

Over-oxygenating a CO₂-retainer can precipitate narcosis (V/Q mismatch + Haldane effect + loss of hypoxic drive). Target 88–92%, not 100%.

3 · Escalate breathing — NIV

NIV (BiPAP) is the highest-impact intervention in acute hypercapnic respiratory failure — it lowers intubation rate and mortality.

Start NIV when	Intubate when
pH ≤ 7.35 with PaCO ₂ > 45	NIV failing / pH still falling
Severe dyspnea, accessory muscle use	Cannot protect airway, ↓consciousness
Persistent hypoxemia despite O ₂	Hemodynamic instability, apnea

GOLD 2026 also positions high-flow nasal cannula as a supportive option, but NIV remains first-line for acute hypercapnic acidosis. Get an early ABG/VBG to find the retainers.

Find the trigger / mimics

- Most common: **viral and bacterial respiratory infection**; also air pollution, nonadherence.
- **Always exclude mimics:** pneumonia (CXR), heart failure (BNP/exam), **PE** (consider in any unexplained exacerbation), pneumothorax, arrhythmia.
- Work-up: CXR, ABG/VBG, ECG, BMP/CBC; viral PCR per season; sputum culture if Pseudomonas risk or vent.

Discharge to prevent the next one · Do/Avoid

- **DO** confirm/optimize **inhaler technique** and step up maintenance: LAMA+LABA; **add ICS** if blood eosinophils elevated and ≥1 moderate/severe exacerbation (GOLD 2026).
- **DO** offer **smoking cessation**, vaccines (influenza, pneumococcal, COVID, RSV), and **pulmonary rehab** referral within 4 weeks (mortality benefit).
- **DO** arrange early follow-up and a written action plan; assess home O₂ need after recovery (not at discharge).
- **AVOID** antibiotics when purulence is absent and not ventilated
- **AVOID** long steroid courses/tapers — 5 days is enough (REDUCE)
- **AVOID** oxygen targets of 94–100% in COPD